भारतीय सूचना प्रौद्योगिकी संस्थान ऊना

(An Institute Of National Importance Under MoE)

Indian Institute of Information Technology Una

School of Electronics

Interim Report - II (April 2026)

Subject Code: ECMC213

Subject Name: Product Development Lab

Academic Year: 2025-26 (Even Semester)

Project Title: ***Interactive Capacitive Touch-Based Multi-LED Control System with Timed Decision Logic and Visual Feedback using ESP32***

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1. PCB Layout Design [FINAL]

Printed Circuit Board (PCB) design is a crucial step in the development of any electronic system, as it provides a compact, reliable, and organized way to connect and support electronic components. In our project, the PCB layout was designed using EasyEDA, keeping in mind proper component placement, routing efficiency, and ease of fabrication.

The figure shows the updated PCB layout of the proposed system. The design is a double-layer PCB consisting of top and bottom copper layers for effective routing. The **red traces represent the top layer (front view) of the PCB,** where most of the signal routing and component connections are visible. The **blue traces represent the bottom layer,** which is used to complete connections, reduce congestion, and improve overall routing efficiency.

The placement of major components such as the ESP32 module, LEDs, resistors, and connectors has been carefully optimized for better accessibility and performance.

For better clarity and understanding, the top view and bottom view of the PCB layout have also been included.

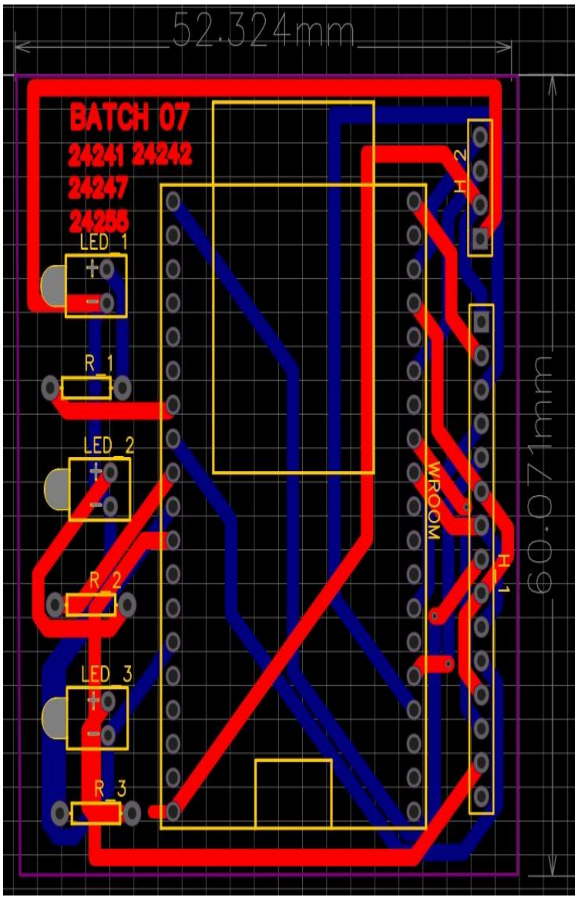

Fig. 1.1 Updated PCB layout design

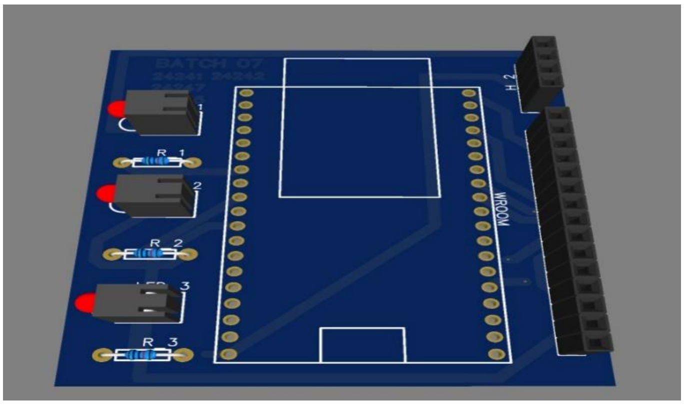

Fig. 1.2 Top View

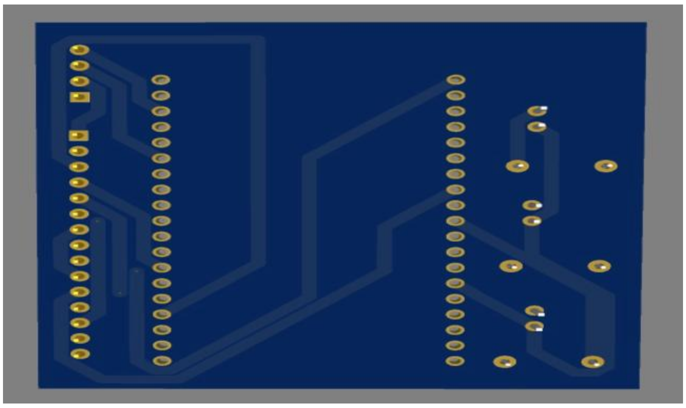

Fig. 1.3 Bottom View

2. 3D Printing Design

For the physical implementation and protection of the circuit, a 3D enclosure box was designed using **SketchUp**, a user-friendly 3D Modeling tool widely used for designing mechanical structures and prototypes. SketchUp provides an intuitive interface that allows precise dimensioning, easy modification, and visualization of 3D objects before fabrication.

The enclosure was designed in a rectangular box shape to accommodate all the electronic components, including the PCB, ESP32 module, and connectors. The dimensions of the box were carefully selected based on the size of the PCB and component placement to ensure a proper fit and sufficient internal space.

According to the design:

- **Length = 8.50 cm**
- **Width = 7.50 cm**
- **Height = 5.00 cm**

The enclosure consists of two parts:

1. **Base (Bottom Section):** Used to hold and support the PCB securely.
2. **Top Cover (Lid):** Designed with rectangular cut-outs to allow access for displays, LEDs, or connectors.

Proper spacing and cut-outs were included in the top section to ensure visibility and accessibility of user interface components such as LEDs and display modules. The design also ensures ease of assembly and maintenance.

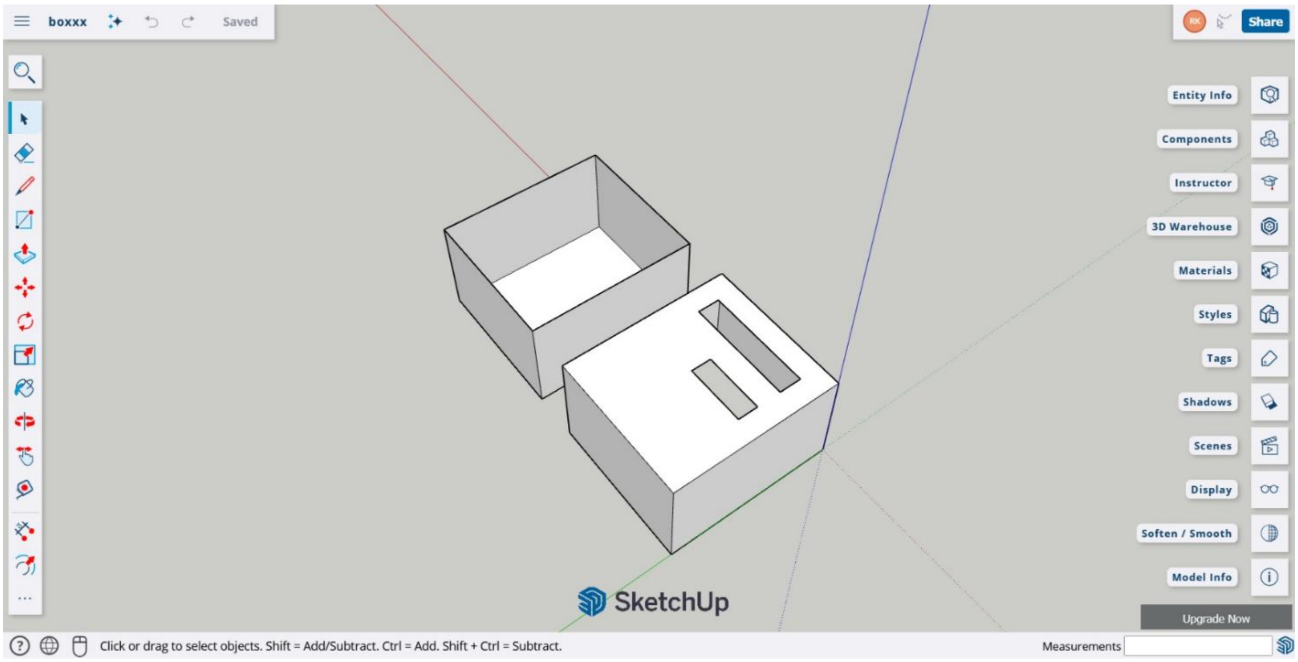

Fig. 2.1 Overall View of 3D Printed Box

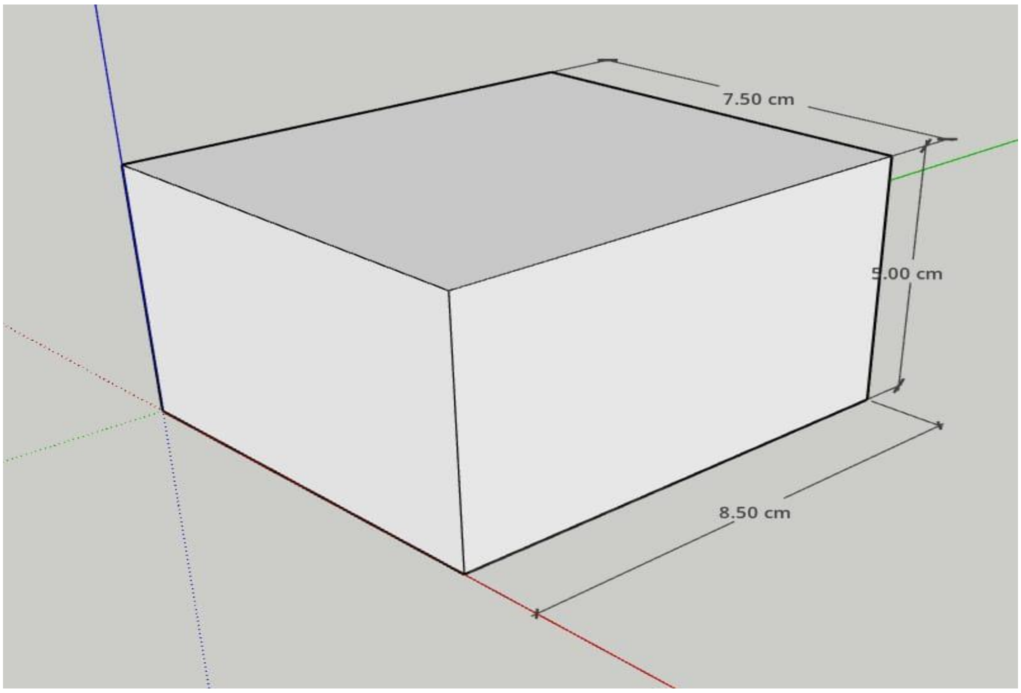

Fig. 2.2 Side View with Dimensions

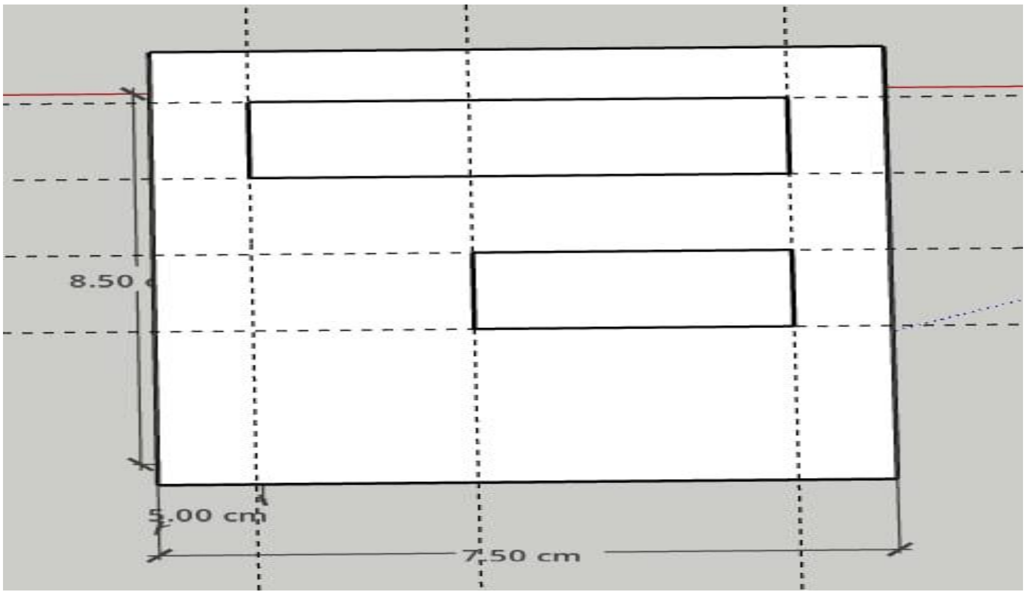

Fig. 2.3 Top View with Dimensions